

Manas Ranjan Dikshit

Keonjhar, Odisha, India | 📞 +91-9337978805 | manasdikshit48@gmail.com

🌐 LinkedIn | 🐙 GitHub | 📄 LeetCode | 📁 Portfolio

Education

B.Tech. in Computer Science & Engineering

Sambalpur University Institute of Information Technology, Burla

CGPA: 9/10

2024 – 2028

Higher Secondary (+2 Science)

KSE, Keonjhar

91%

2022

Skills

Programming: Python, Java, C, SQL, JavaScript, TypeScript

AI & ML: Machine Learning, LLM Applications, Langchain, AI Agents, Generative AI, MCP, RAG

Frameworks: Django, Spring Boot, React, Next.js, Tailwind CSS, Angular.js

Systems: Docker, Kubernetes, GitHub Actions, CI/CD, Distributed Systems, REST APIs

Databases: PostgreSQL, MySQL, Supabase, PostGIS, Firebase, VectorDB

Experience

Open-Source Contributor

Dec 2024 – Present

Apache, CNCF, Kubernetes, Webpack, LibreOffice, DBpedia,

OSGeo, Checkstyle, OpenStreetMap, cBioPortal, AsyncAPI

Python | Java | Distributed Systems

- Contributed to **15+ global open-source organizations** including Apache, CNCF, Kubernetes, Webpack, LibreOffice, DBpedia, and OSGeo, with **1700+ commits** across 2025–2026.
- Developed and merged **1600+ LOC** enhancements spanning **Maven Compiler Plugin**, **Webpack**, **Thread Loader**, schema validation, API robustness, null safety, annotation processing, and enterprise-scale frontend tooling.
- Performed **debugging**, **root-cause analysis**, **performance optimization**, and improved **CI/CD**, testing, caching, automation, security, and reliability across **distributed systems**, **cloud-native**, **AI/ML**, and **geospatial** projects.
- Collaborated with global maintainers through **code reviews**, technical discussions, issue triage, documentation, and iterative PR refinements across multiple open-source ecosystems.

Projects

AQI-HCHO Prediction Model – ISRO 2026 🌐

- Developed an **end-to-end ML/GIS pipeline** for **surface AQI prediction** and **HCHO hotspot detection** over India using **TROPOMI**, **INSAT-3D**, **ERA5**, **CPCB**, and **NASA FIRMS** datasets.
- Built and evaluated **Random Forest**, **Gradient Boosting**, **CNN-LSTM**, and **ConvLSTM** models with advanced **spatio-temporal feature engineering** for environmental forecasting.
- Implemented **DBSCAN hotspot clustering**, anomaly detection, GeoJSON export, automated CLI workflows, and a **7-page Streamlit dashboard** for interactive geospatial analytics.
- Integrated **PyTorch**, **Scikit-learn**, and **Streamlit** to enable scalable model training, visualization, and end-to-end environmental monitoring on a **0.1° India-wide grid**.

Messenger Automation – Automated Birthday SMS System 🌐

- Engineered a **Python 3.12** automation system integrating **GitHub Actions**, **Android SMS Gateway REST API**, and **CSV-based** contact management to schedule personalized SMS delivery directly through a user's SIM card.
- Implemented **template rendering**, **delivery polling**, **persistent state management**, **duplicate-send prevention**, **exponential backoff**, and **timezone-aware scheduling** for reliable message orchestration.
- Built a production-ready workflow with **PyTest**, **GitHub CI/CD**, **GitHub Secrets**, structured logging, environment-based configuration, and comprehensive documentation for maintainable and secure deployment.

ResQAI – Community-Validated Federated Disaster Intelligence 🌐🌐

- Developed an **AI-powered disaster intelligence platform** with **React**, **TypeScript**, **Vite**, **Tailwind CSS**, **Leaflet.js**, **PostgreSQL**, **Supabase**, and **Deno Edge Functions** for **real-time risk prediction**, community validation, and emergency response.
- Built an **offline-first PWA** featuring **interactive geospatial heatmaps**, **SOS geolocation**, **AI chatbot**, multilingual (**English**, **Hindi**, **Odia**) support, explainable AI, and live integrations with **USGS**, **NASA EONET**, **Open-Meteo**, and **OSRM APIs**.
- Implemented **role-based authentication**, **real-time WebSocket updates**, **TanStack Query**, **Framer Motion**, **i18next**, and **React Hook Form + Zod** to deliver scalable disaster monitoring across a **15-point geospatial prediction grid** with **6 disaster risk categories**.

VoiceUp India – AI-Powered Citizen Engagement Platform 🌐🌐

- Built a **full-stack civic engagement platform** using **React**, **TypeScript**, **Vite**, **Tailwind CSS**, **Supabase**, **PostgreSQL**, **PostGIS**, and **React-Leaflet** for reporting, tracking, and analyzing public issues with geospatial visualization.
- Implemented **AI-powered civic assistance**, **real-time dashboards**, **interactive maps**, **commenting**, **upvote/downvote**, notifications, and location-aware issue correlation using **PostGIS**, **TanStack Query**, and **Google Maps APIs**.
- Developed secure backend services with **Supabase Functions**, **Row-Level Security (RLS)**, file attachments, and real-time synchronization, enabling transparent citizen-government collaboration through scalable cloud architecture.

Awards & Achievements

- Hackathon Winner:** Hack4Impact 2025, HackVerse 2025, Hack-O-thor 2024, HackVerse 2.0 2026; **President, AWS Student Builders Club (SUIIT)**; **State Startup Summit Winner (2025)**; **Apache Open-Source Contributor**; **National Debate Champion (2022)**; **Google Cloud Skill Boost Top Performer (2024)**; **District Topper (10th & 12th Boards)**.